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Urban green space percolation: From connectivity to ecological management mapping

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ABSTRACT

Urban green spaces (UGS) are increasingly planned as connected networks, yet many connectivity assessments reduce cost variation to a static threshold and provide limited guidance on where to intervene. We develop a percolation-based framework for translating dynamic UGS connectivity into ecological management priorities. The framework constructs a least-cost, cost-weighted UGS network, activates edges in ascending cost-weighted distance to identify critical transitions and merger hierarchies, and converts the full percolation history into management zones, a three-level edge hierarchy, and a four-type node typology. Applied to Nanjing, China, the analysis shows that system-level connectivity emerges within a narrow high-cost critical band triggered by a small set of cross-barrier links. The merger history reveals three persistent spatial subsystems, which are further resolved into five management zones. Robustness experiments indicate that protecting map-prioritized edges and nodes slows the decline of weighted global efficiency under the tested removal rules. The framework links process-based connectivity diagnosis with spatially explicit planning actions, including corridor safeguarding, within-zone reinforcement, and stepping-stone protection, while remaining transferable with local calibration.

1. Introduction

Urban green spaces (UGS) are widely recognized as cornerstones of urban sustainability (Wolch et al., 2014). They are core components of blue–green infrastructure (BGI), understood here as connected parks, ecological corridors, waterways, wetlands, rain gardens, and other permeable or vegetated spaces. Through infiltration and temporary storage, these spaces help reduce runoff, filter pollutants, support groundwater recharge, and moderate erosion; together with biodiversity support and microclimate regulation, these functions sustain ecological flows across urban landscapes (Aronson et al., 2017; Fang et al., 2023; Heidt & Neef, 2008; Molné et al., 2023). Urban expansion, however, increasingly fragments these assets through impervious surfaces, transport corridors, and other high-resistance land uses (Li et al., 2019; Zou et al., 2025). Their ecological value therefore depends not only on the area or quality of individual patches, but also on whether patches remain connected as a coherent system capable of supporting movement, dispersal, and related ecological processes (Taylor et al., 1993; Wu et al., 2020).

Connectivity assessment has become central to UGS planning, but many outputs remain difficult to translate into differentiated management actions. Landscape metrics, graph measures, and resistance-based models describe important aspects of spatial structure and movement cost, while least-cost path (LCP) analysis provides interpretable corridor representations (Adriaensen et al., 2003; McRae, 2006; Saura & Pascual-Hortal, 2007; Urban & Keitt, 2001). Yet static or single-threshold outputs can obscure how connectivity emerges across increasing cost levels and which network elements govern the transition from fragmented local clusters to a system-spanning structure. Percolation theory is well suited to reveal this cost-ordered assembly process, but its use in urban connectivity studies has more often emphasized thresholds and vulnerability than management-ready spatial categories (Behnisch et al., 2019; Bian et al., 2023; Habrich & Fahrig, 2025).

This study develops an integrated framework that operationalizes percolation theory on a least-cost, cost-weighted UGS network. The framework identifies the critical band where system-level connectivity

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emerges, reconstructs the associated merger hierarchy, and translates the full percolation history into ecological zones, edge priorities, and node roles. Applied to Nanjing, China, it provides a case-based test of how process-based connectivity diagnosis can support spatially explicit ecological management and how the resulting priorities perform under robustness experiments based on weighted global efficiency.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature and positions the study. Section 3 presents the methodological framework. Section 4 applies the framework to Nanjing and reports the percolation, mapping, and robustness results. Section 5 discusses the planning implications, main insights, limitations, and future research directions. Section 6 concludes.

2. Literature review

This section situates the study within existing approaches to UGS connectivity assessment and clarifies the practical gap that motivates a percolation-based management-mapping workflow.

2.1. Main pathways for assessing UGS connectivity

UGS connectivity has been evaluated through several complementary analytical traditions. Landscape metrics summarize patch size, shape, isolation, and fragmentation, thereby providing compact descriptions of spatial configuration. Graph-theoretic approaches represent patches as nodes and connections as edges, enabling network-level metrics such as centrality, component structure, and habitat availability (Saura & Pascual-Hortal, 2007; Saura & Torné, 2009; Urban & Keitt, 2001). Resistance-based methods go further by embedding landscape heterogeneity into connectivity assessment through movement cost or permeability surfaces (Adriaensen et al., 2003; Dutta et al., 2022; McRae, 2006). Recent urban studies have extended these approaches through high-resolution green-space datasets, species-oriented LCP models, and comparisons between LCP and circuit-theory outputs (Aleixo et al., 2024; Kwon et al., 2021; MacKinnon et al., 2023). Together, these traditions have substantially advanced how urban ecological structure is measured, compared, and interpreted, but they most often remain diagnostic rather than explicitly prescriptive for planning.

2.2. Strengths and limitations of LCP and related cost-based connectivity methods

Among resistance-based approaches, LCP has been especially influential because it incorporates ecological heterogeneity into connectivity modeling and produces corridor representations that are readily interpretable for planning. Related cost-based methods can reveal connections that are invisible under purely Euclidean or binary-threshold approaches. At the same time, LCP analysis has important limitations. It emphasizes one least-cost route rather than multiple plausible pathways, so it can underrepresent redundancy, diffuse movement, and area-based flow (Hall et al., 2021; McRae et al., 2008). Its outputs are also sensitive to the construction of the resistance surface, including data source choice, variable weighting, and ecological assumptions (Koen et al., 2012; Savary et al., 2022). Moreover, cost-based analyses are often converted back into static snapshots at one or a few thresholds, which obscures how connectivity assembles, stabilizes, or fails across the cost gradient. Even when such methods identify important corridors, they do not by themselves define management zones, intervention hierarchies, or differentiated functional roles for network elements (Balbi et al., 2021; Habrich & Fahrig, 2025).

2.3. Percolation as a bridge from diagnosis to management mapping

Percolation analysis introduces a process perspective by ordering connections along a control variable and tracking how local clusters merge into a system-spanning structure (Breskin et al., 2006). In ecological and spatial networks, this makes it possible to identify critical transition bands, persistent subsystems, and hierarchical merger sequences rather than evaluating connectivity at a single arbitrary threshold. Recent percolation studies, simulation-based urban green-space models, and urban connectivity reviews have respectively highlighted critical transitions, hard-to-detect corridors, and validation gaps, but these insights have usually remained descriptive: they identify transitions, thresholds, or sensitivities without converting them into operational planning priorities (Behnisch et al., 2019; Bian et al., 2023; Dong et al., 2025; Habrich & Fahrig, 2025).

This literature points to a translational gap. Existing methods can diagnose patch structure, movement cost, and connectivity transitions, but planning decisions require outputs that distinguish where to protect cross-zone backbones, where to reinforce internal redundancy, and which patches play distinct functional roles. As summarized in Table 1, the key limitation is not the absence of connectivity metrics, but the limited conversion of dynamic connectivity evidence into integrated management categories. A percolation-based management workflow can address this gap when the full assembly history is retained and translated into zones, edge hierarchy, and node typology rather than treated only as a threshold-detection exercise (Adriaensen et al., 2003; Bian et al., 2023; Habrich & Fahrig, 2025; Urban & Keitt, 2001).

3. Methodology

We develop an integrated, three-stage framework that operationalizes percolation theory for UGS management, as illustrated in Fig. 1. First, we construct a cost-weighted UGS network in which patches are represented as nodes and their connections as ecologically meaningful LCPs weighted by cost-weighted distance (CWD). This network encodes potential ecological flow pathways and their relative difficulty. Second, we run a percolation scan that incrementally activates edges in order of increasing cost to reconstruct the dynamic assembly of the network and to identify connectivity transitions and the associated emergence hierarchy. Finally, we translate these process-based insights into a spatially explicit management map by classifying network elements into process-defined ecological zones, a prioritized hierarchy of connective edges, and a node typology that reflects their distinct contributions to system-level connectivity. We additionally perform robustness experiments under edge and node removal to test whether the resulting process-based priorities identify structurally critical components when evaluated in terms of cost-based functional connectivity.

3.1. Construction of the UGS network

The first stage constructs a cost-weighted UGS network in three steps: (i) delineating UGS patches, (ii) modeling a composite resistance surface, and (iii) computing LCPs between patches (Fig. S1, Step 1). Nodes represent patches, and edges represent LCPs weighted by CWD.

3.1.1. UGS patch extraction

UGS patches are delineated from land cover data. To reduce noise from ecologically marginal fragments and to keep the network computationally tractable, we apply an area-based screening. A threshold derived from the patch area–frequency distribution is used to remove very small patches that contribute little to functional connectivity (Mokhtari et al., 2022). The retained patches are then designated as the nodes of the network.

Table 1
Comparative positioning of representative connectivity approaches. Classical and recent studies provide a strong basis for connectivity diagnosis, but most stop short of translating process information into spatially explicit management priorities.

Representative references	Approach	Cost and dynamics	Planning contribution and remaining gap
Urban and Keitt (2001); Saura and Pascual-Hortal (2007)	Landscape and graph metrics	Cost is indirect or simplified; dynamics are not explicit.	Establishes diagnostic measures of patch and network structure, but spatial intervention logic usually requires additional interpretation.
Adriaensen et al. (2003); McRae (2006)	Cost-based connectivity	Resistance is explicit; outputs are usually evaluated as static routes or thresholds.	Represents heterogeneous movement costs and plausible routes, but management zones and intervention hierarchies are not directly specified.
Balbi et al. (2021); Aleixo et al. (2024); Kwon et al. (2021)	Resistance and comparative analysis	Cost surfaces and scenarios are explicit; dynamic emergence is usually not reconstructed.	Improves ecological realism and comparative assessment, but results often remain descriptive or scenario-specific for planning.
Bian et al. (2023); Dong et al. (2025)	Percolation and process-oriented connectivity	Cost or thresholds can be ordered to reveal transitions and emergence.	Reveals thresholds, sensitivities, or difficult-to-prioritize corridors, but translation into integrated management categories remains limited.

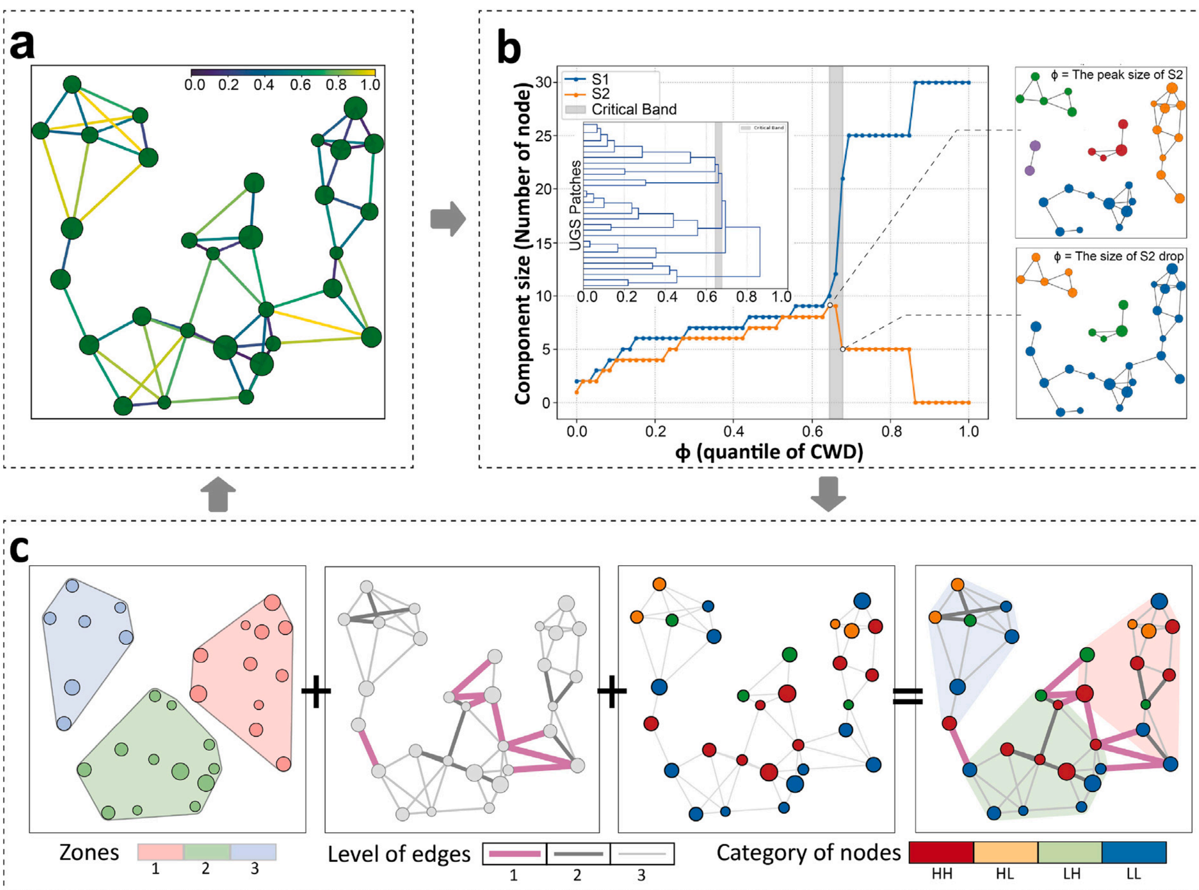

Fig. 1. Conceptual illustration of the methodological framework. The framework translates percolation theory into an actionable management map by systematically moving from connectivity diagnosis to mapping. **a,** Cost-weighted UGS network: Nodes are UGS patches and edges are LCPs; edge color encodes quantile of CWD (ϕ). **b,** Percolation Scan: Trajectories of the largest (S_1) and second-largest (S_2) components reveal a narrow critical band (gray band) where S_2 peaks and then collapses into S_1; right insets show network states at the S_2 peak and drop; embedded dendrogram depicts the merger hierarchy. **c,** Ecological Management Mapping: Insights gathered from the full percolation scan are systematically translated into a multi-level management map by synthesizing management zones, prioritized edges, and functional nodes.

3.1.2. Resistance surface modeling

We model a composite resistance surface to quantify the ecological cost of movement across the landscape (Dutta et al., 2022). The surface synthesizes multiple natural and anthropogenic factors that influence ecological flow, such as land cover type, vegetation status, topography and proximity to infrastructure (Koen et al., 2012; Li et al., 2023; Sun et al., 2024). To derive objective weights for these factors, we employ spatial principal component analysis (SPCA) (Li et al., 2024) and combine standardized factor layers into a single resistance raster, in which lower values indicate higher permeability and higher values denote stronger barriers to movement. In this representation, resistance can be interpreted as the instantaneous difficulty faced by ecological flows when traversing each cell.

3.1.3. Least cost paths calculation

Using the UGS patches as nodes and the resistance surface as the cost layer, we compute LCPs between all node pairs with the Linkage Mapper toolbox (McRae & Kavanagh, 2011; Sawyer et al., 2011). The weight of each edge in the resulting network is its CWD, defined as the minimum cumulative resistance incurred along the LCP (Savary et al., 2022). Ecologically, CWD represents the least effort required for an organism or ecological flow to move between two patches through the surrounding landscape. This procedure yields an undirected, cost-weighted graph that serves as the empirical basis for the subsequent percolation analysis and robustness experiments. Formal definitions of path cost, LCP and CWD are provided in supplementary material, S1.1.

3.2. Percolation scan: Revealing connectivity transition and structural emergence

In the second stage, we apply a percolation scan to the cost-weighted UGS network to reveal its dynamic assembly and structural hierarchy (Fig. S1, Step 2). As summarized in Fig. 1b, we track the evolution of the largest (S_1) and second-largest (S_2) components during this process. This treats connectivity as a dynamic bottom-up formation process rather than as a static snapshot.

The scan simulates network growth by progressively adding edges in ascending order of their CWD. To establish a scale-invariant control parameter, we transform CWD values into empirical quantiles $\phi \in [0, 1]$. This ordering provides a clear ecological interpretation: low ϕ values correspond to permeable corridors, while high ϕ values represent resistant pathways that traverse dense urban fabric, major rivers, or other strong barriers. As we increase ϕ, disconnected patches merge, and we track the sizes of the S_1 and S_2 components (Dorogovtsev et al., 2008; Goltsev et al., 2003). We define the percolation transition as the interval where S_2 peaks and subsequently collapses into S_1, marking the rapid emergence of a system-spanning giant component (Achlioptas et al., 2009; Fan et al., 2020). This transition signifies the qualitative shift from a fragmented landscape to an integrated ecological network.

Finally, we summarize the complete multi-scale assembly process in a hierarchical emergence dendrogram, conceptually illustrated as an inset in Fig. 1b. This dendrogram encodes the entire bottom-up merger history, tracking how individual patches coalesce into growing clusters and, ultimately, into the global network. This compact representation defines the network's structural hierarchy and underpins our subsequent management classification.

3.3. Ecological management mapping: Integrating zones, edges, and nodes

In the final stage, we translate the percolation results into a spatially explicit management strategy (Fig. S1, Step 3). We classify network elements into three linked layers: management zones, a prioritized edge hierarchy, and a functional node typology (Fig. 1c). Collectively, these layers identify leverage points for protecting, reinforcing, or restoring system-level connectivity under cost constraints. Fig. 2 expands this final step into an operational workflow, showing how the zone-delineation matrices are assembled, how edge hierarchy is derived, and how node typology is assigned.

3.3.1. Zone: Geographically regularized consensus clustering

To delineate macroscopic ecological zones that are both functionally cohesive and spatially coherent, we use geographically regularized consensus clustering (Figs. 1c; 2a). This approach synthesizes the network's dynamic functional topology, captured across the entire percolation process, with its static spatial structure defined by ecological proximity. In practical terms, the zoning step combines patches that repeatedly function together with the cost-based proximity needed for coherent spatial management. The method proceeds in three steps.

First, we construct a functional consensus matrix **A** to distill stable, multi-scale co-occurrence patterns of patch pairs (Lancichinetti & Fortunato, 2012; Shi et al., 2022). Unlike a static snapshot at an arbitrary threshold, this matrix integrates information across the full range of connectivity scales. Each element A_{ij} quantifies the proportion of percolation steps in which patches i and j belong to the same connected component, thus measuring their persistent functional association:

$$A_{ij} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \mathbf{1}\big(S_i(p) = S_j(p)\big), \tag{1}$$

where $\mathcal{P}$ is the set of discrete cost-quantile steps, $S_i(p)$ is the cluster ID (connected-component label) of patch i at step p, and $\mathbf{1}(\cdot)$ is the indicator function. The resulting matrix **A** represents the network's intrinsic functional topology, but it is agnostic to geography and can group distant patches.

Second, to impose spatial coherence, we derive an ecological proximity matrix, **K**, based on the CWD between patch pairs. We transform the CWD matrix, $\mathbf{D}_{\mathrm{cwd}}$, into a normalized similarity matrix **K**, where higher values indicate greater proximity:

$$K_{ij} = 1 - \frac{D_{\mathrm{cwd}}[i,j] - \min(D_{\mathrm{cwd}})}{\max(D_{\mathrm{cwd}}) - \min(D_{\mathrm{cwd}})}. \tag{2}$$

This matrix serves as a geographic regularizer, encouraging the final zones to be spatially contiguous and ecologically interpretable.

Finally, we fuse the functional and spatial matrices into a single hybrid similarity matrix, **M**, through a linear combination controlled by a regularization parameter $\lambda \in [0, 1]$:

$$\mathbf{M} = (1 - \lambda) \cdot \mathbf{A} + \lambda \cdot \mathbf{K}. \tag{3}$$

This formulation establishes a tunable trade-off between functional association (matrix **A**) and spatial proximity (matrix **K**). The value of λ is selected based on sensitivity analysis that balances cluster quality and spatial coherence; its case-specific choice is reported in the application section. Spectral clustering is then applied to matrix **M**, a method well suited to detecting non-convex clusters in geographic data (Ding et al., 2024; Ng et al., 2001). The number of clusters (zones) is determined using the eigengap heuristic (Von Luxburg, 2007), yielding a final partition that is functionally robust and spatially explicit. The resulting management zones provide the spatial units within which edge levels and node roles are interpreted.

3.3.2. Edges: A synthesis of functional role and process robustness

At the edge level, we assess the importance of each connection using a dual-criteria framework that combines its topological role with its dynamic persistence during network formation. This framework classifies all edges into a three-tier hierarchy, providing a prioritized guide for management (Figs. 1c; 2b).

The first criterion, functional role, distinguishes edges by their position relative to the ecological zones. An edge is classified as inter-zonal if it connects patches assigned to different zones, and as intra-zonal if it connects patches within the same zone.

The second criterion, process robustness, quantifies the dynamic stability of intra-zonal edges over the percolation process. We introduce a process robustness index, R_{ij}, which measures an edge's average structural prominence throughout the scan (Shen et al., 2021). It is calculated as the mean relative size of the component to which the edge belongs, averaged across all percolation steps where the edge is active:

$$R_{ij} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \frac{|S_i(p)|}{|S_{\max}(p)|} \cdot \mathbf{1}\big(S_i(p) = S_j(p)\big), \tag{4}$$

where $|S_i(p)|$ is the size of the cluster containing nodes i and j, and $|S_{\max}(p)|$ is the size of the largest cluster at step p. A high R_{ij} score indicates that an edge belongs to the early-forming and persistent structural component, and is therefore robust across multiple connectivity scales.

These two criteria are integrated to establish the final three-tier hierarchy:

- **Level 1:** All inter-zonal edges, interpreted as cross-zone backbones of the whole network because they bridge distinct ecological zones and maintain city-wide ecological exchange.

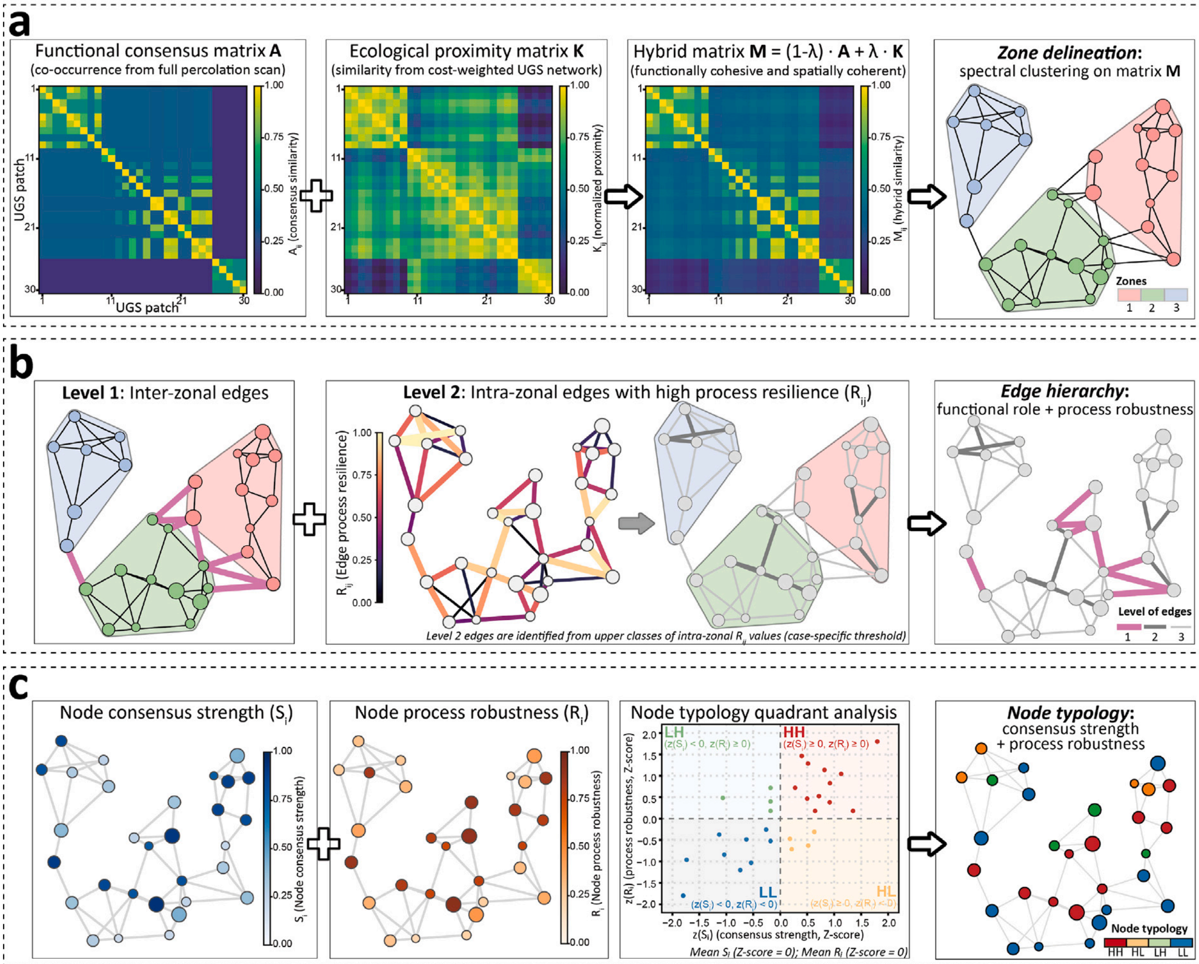

Fig. 2. Operational workflow for ecological management mapping. a, Zone delineation combines the functional consensus matrix **A**, derived from co-occurrence across the full percolation scan, with the ecological proximity matrix **K**, derived from the cost-weighted UGS network. The hybrid matrix $\mathbf{M} = (1-\lambda)\mathbf{A} + \lambda\mathbf{K}$ is then used for spectral clustering to obtain spatially coherent management zones. **b,** Edge hierarchy integrates inter-zonal functional role with intra-zonal process robustness R_{ij} to distinguish cross-zone backbones from robust and residual within-zone edges. **c,** Node typology combines consensus strength S_i and process robustness R_i in standardized space, yielding four relative functional roles.

- **Level 2:** The subset of intra-zonal edges with high process robustness (R_{ij}), forming the robust connective core within each zone.
- **Level 3:** All remaining intra-zonal edges, which provide fine-grained local permeability and redundancy.

The distinction between Level 2 and Level 3 is case-specific. In the Nanjing application, it is derived from the upper classes of the empirical R_{ij} distribution rather than imposed as a case-independent ecological threshold.

3.3.3. Nodes: A synthesis of consensus strength and process robustness

At the node level, we determine functional roles using a complementary dual-criteria framework that combines structural cohesiveness with dynamic prominence, evaluated over the entire percolation process. This yields a four-part typology that classifies each node according to its contribution to overall network integrity (Fig. 1c; Fig. 2c).

The first criterion, consensus strength (S_i), quantifies a node's structural coreness and clarity of belonging across connectivity scales. It measures the extent to which a node consistently co-clusters with other nodes throughout the percolation process. Specifically, S_i is defined as the average co-membership strength with all other nodes, derived from the consensus matrix **A**:

$$S_i = \frac{1}{N-1}\sum_{j\neq i} A_{ij}. \tag{5}$$

A high S_i indicates that the node is a core member of a persistent, tightly knit group.

The second criterion, process robustness (R_i), measures a node's dynamic prominence across the percolation scan. It is computed as the mean relative size of the component to which the node belongs, averaged across all percolation steps:

$$R_i = \frac{1}{|\mathcal{P}|}\sum_{p\in\mathcal{P}} \frac{|S_i(p)|}{|S_{\max}(p)|}. \tag{6}$$

Nodes with high R_i join large structural components early and remain within them over a broad range of cost levels.

To classify nodes into functional roles, we standardize both S_i and R_i to z-scores within the focal network. In this standardized space, zero corresponds to the network-wide mean. The zero boundary is therefore used as a parsimonious within-system split between above-mean and below-mean roles, with the high side defined as values greater than or equal to zero and the low side as values below zero. The resulting partition is a relative, parameter-light role classification rather than an absolute ecological cutoff or a case-independent threshold transferable without local calibration.

A quadrant analysis based on these standardized S_i and R_i scores assigns each node to one of four functional types:

- **HH** ($z(S_i) \geq 0,\ z(R_i) \geq 0$): Nodes that are consistently embedded in large, cohesive components throughout the percolation process. They form stable cores of ecological zones and serve as persistent, high-capacity hubs for ecological flow.

- **LH** ($z(S_i) < 0,\ z(R_i) \geq 0$): Nodes that frequently belong to large components but exhibit weak and shifting co-membership patterns. They typically lie at interfaces between zones and function as dynamically important stepping-stones that enable long-range or cross-zone connectivity.
- **HL** ($z(S_i) \geq 0,\ z(R_i) < 0$): Nodes that are strongly integrated within a locally cohesive group but remain confined to relatively small components over much of the scan. They are important for maintaining internal cohesion and redundancy within a zone, yet play a limited role in early or system-spanning connectivity.
- **LL** ($z(S_i) < 0,\ z(R_i) < 0$): Nodes that rarely join large or persistent components and show weak, unstable co-membership across steps. They occupy peripheral or fragmented positions and contribute little to the formation or maintenance of overall connectivity.

This typology provides a process-aware basis for distinguishing leverage points, structural anchors, and vulnerable components in subsequent management analysis.

3.4. Robustness experiments

To assess whether the percolation-based classifications capture structurally important components of the UGS network, we perform robustness experiments under progressive removal of edges and nodes. These experiments are intended as comparative structural tests rather than direct validation of realized species movement: they ask whether components highlighted by the ecological management map have disproportionate effects on cost-based system-level connectivity under the adopted network representation. We use weighted global efficiency, E, as the response variable:

$$E = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{1}{D_{\mathrm{cwd}}[i, j]}, \tag{7}$$

where $D_{\mathrm{cwd}}[i, j]$ is the minimum cumulative resistance incurred along the LCP between nodes i and j. E is normalized to 1 for the intact network. Because E decreases as paths become longer or more circuitous in cost terms, it reflects how easily ecological flows can be routed through the network at the system level.

We compare five removal rules for both edges and nodes (Table 2). For edges, Map-attack removes inter-zonal edges first and then intra-zonal edges, with each group ordered by descending edge process robustness R_{ij}; Map-protect applies the reverse logic. For nodes, Map-attack follows the typology order HH → LH → HL → LL, whereas Map-protect follows LL → HL → LH → HH. The static-topology benchmark uses weighted betweenness centrality. The ecological-function benchmark uses an edge gravity score and a node-level dPC score computed on the intact cost-weighted graph. Random removal is repeated 50 times, with the mean trajectory and the 10%–90% interval reported.

These benchmarks represent established topological and ecological-function alternatives rather than exact one-to-one replicas of directly comparable management-mapping frameworks. For each curve, we summarize the response using three complementary metrics: the area under the normalized efficiency curve (AUC_E), the remaining efficiency after 20% of components have been removed ($E@20\%$), and the removal fraction at which E first falls below 0.5 ($f_{E<0.5}$). Higher values indicate a slower decline of connectivity for a given removal rule. A detailed description of the benchmark measures used for the static-topology and ecological-function comparisons is provided in supplementary material, S1.2.

4. Case study and results

We apply the proposed framework to Nanjing, China, to demonstrate how a cost-ordered percolation perspective can be translated into management-ready spatial priorities for UGS connectivity. Section 4.1 introduces the study area and datasets. Section 4.2 constructs the cost-weighted UGS network and summarizes its key characteristics. Section 4.3 reports the percolation transition and the associated emergence hierarchy across scales. Section 4.4 presents the resulting ecological management mapping for Nanjing (zones, edge hierarchy, node typology, and the integrated map). Finally, Section 4.5 evaluates the robustness of the percolation-based priorities under progressive edge and node removal.

4.1. Study area and data

Our case study focuses on Nanjing, the capital of Jiangsu Province in eastern China (Fig. 3a). As a major Yangtze River Delta metropolis and regional economic, industrial, and transport hub, Nanjing has experienced rapid urban expansion, infrastructure intensification, and continuing pressure on green and blue spaces (Li et al., 2024; Shen et al., 2025). Expanding impervious surfaces and transport barriers have intensified fragmentation across its mountain–river–wetland BGI system. Relatively intact green concentrations occur in the south, east, and west, while the metropolitan core, major transport corridors, and the Yangtze River separate green subsystems across a dense urban matrix. This barrier-structured landscape provides an appropriate case for evaluating whether a cost-ordered percolation framework can distinguish local cohesion, cross-barrier integration, and structurally decisive corridors within the same urban network.

We use a suite of geospatial datasets to delineate UGS patches and to build the comprehensive resistance surface. Land cover data provide the basis for patch extraction, while six factors represent landscape resistance: land-cover type, normalized difference vegetation index (NDVI), distance to railways, distance to major roads, nighttime light intensity, and slope derived from a digital elevation model (DEM). Detailed descriptions of the datasets are summarized in Table S5.

4.2. Cost-weighted UGS network construction

The cost-weighted UGS network for Nanjing is constructed following the procedure in Section 3.1. First, we extract UGS patches (forest, shrub, and wetland classes) from the land cover data using an 8-neighbor connectivity rule. The initial extraction yields 5184 patches, with nearest-neighbor distances concentrated within several tens to just over one hundred meters (Fig. S2a). To reduce noise from very small fragments and to keep the analysis computationally tractable, we apply area-based screening using the patch area–frequency distribution (Fig. S2b). This results in 500 retained patches that still account for 93.9% of the total UGS area (Fig. 3b).

Second, we generate a comprehensive resistance surface by integrating the six resistance factors introduced above (Table S5). To avoid subjective weighting, we employ SPCA to derive data-driven weights for each factor. The standardized factor layers are input to the SPCA, and the resulting component loadings and variance contributions are used to compute the final weights. The procedure and the reclassification scheme for each factor are documented in Tables S6 and S7. The reclassified resistance layers are then combined via a weighted sum to produce the resistance surface shown in Fig. 3c.

Finally, treating the screened UGS patches as nodes and the resistance surface as the traversal cost layer, we use the Linkage Mapper toolbox to generate LCPs among patches. Each linkage is represented as an edge and assigned the CWD equal to the minimum cumulative resistance along its LCP. The resulting network contains 500 nodes and 1179 edges and is connected, providing the empirical basis for the percolation analysis (Fig. 3d).

Table 2
Removal rules used in the robustness experiments. All rules are applied separately to edge removal and node removal, and all responses are measured using normalized weighted global efficiency E.

Rule	Operational definition	Purpose of comparison
Map-attack	Removes components in the priority order implied by the ecological management map. Edges follow inter-zonal before intra-zonal ordering, ranked by R_{ij} within groups. Nodes follow HH → LH → HL → LL.	Tests whether map-prioritized components are structurally vulnerable leverage points whose loss rapidly reduces system-level connectivity.
Map-protect	Removes components in the reverse map order. Edges postpone inter-zonal and high-R_{ij} links. Nodes postpone HH and LH nodes.	Tests whether safeguarding map-prioritized components delays connectivity loss under progressive disturbance.
Static topology	Removes components by descending weighted betweenness centrality on the intact cost-weighted graph.	Provides a classical graph-theoretic benchmark based on shortest-path mediation.
Ecological function	Removes edges by descending gravity score and nodes by descending dPC, both computed on the intact cost-weighted graph.	Provides a comparator based on size-cost interaction for edges and probability-of-connectivity contribution for nodes.
Random mean	Removes edges or nodes in random order over 50 repeated runs; the mean curve and the 10%–90% interval are summarized.	Provides a null benchmark for non-targeted disturbance.

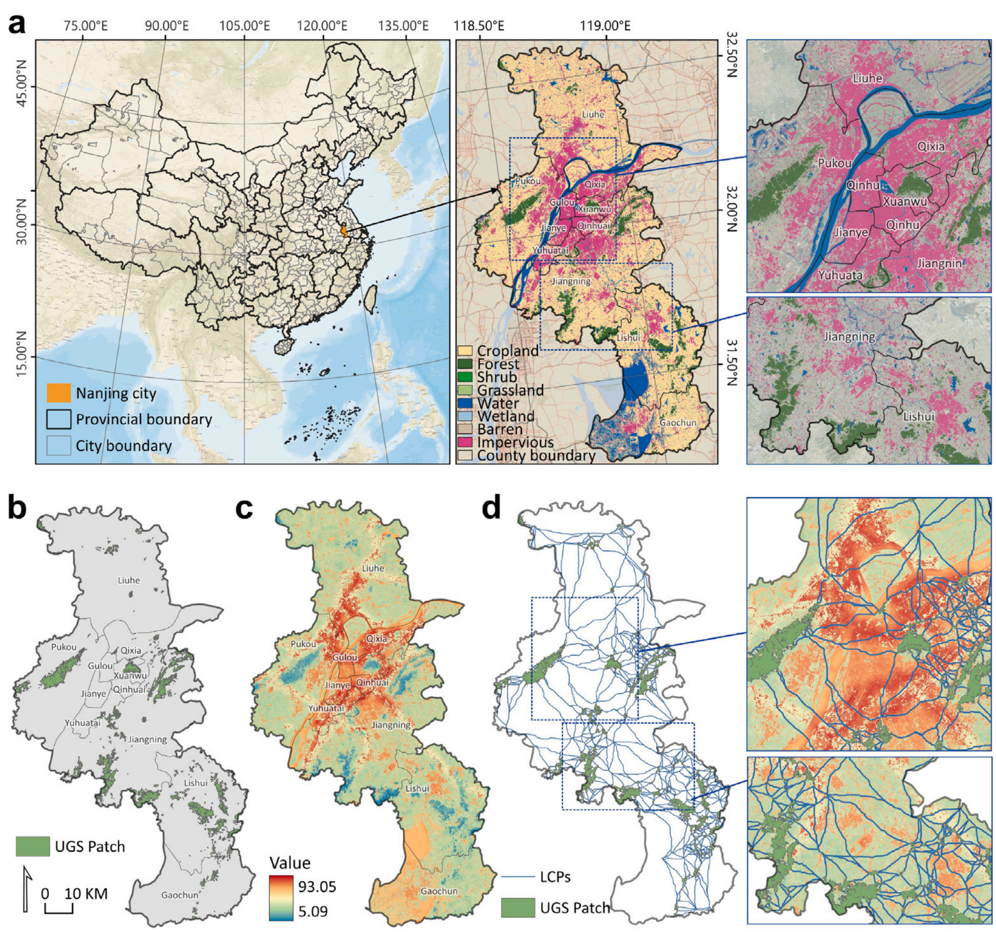

Fig. 3. Study area and the UGS network construction. a, Location of Nanjing, China, and the land cover context. The insets highlight the two primary geographical barriers defining the landscape: the high-resistance urban area (impervious, pink) and the Yangtze River (blue). **b,** The extracted UGS patches (nodes) for analysis. **c,** The comprehensive resistance surface, where warmer colors indicate higher resistance to ecological flow. **d,** The final cost-weighted UGS network forms the empirical basis for the percolation analysis.

4.3. Percolation scan

We implement the percolation scan on the Nanjing cost-weighted UGS network by progressively adding edges in ascending order of their CWD. Prior to the scan, all CWD values are converted to empirical quantiles, $\phi \in [0,1]$, to obtain a normalized cost scale that is comparable across steps (Fig. S4). The main percolation outcomes are summarized in Fig. 4, including the evolution of the S_1 and S_2, the incremental gains of S_1 in count share and area share, and the associated emergence dendrogram capturing the merger hierarchy.

4.3.1. Percolation transition

Fig. 4a shows the trajectories of S_1 and S_2 as ϕ increases. Across most of the cost range, S_1 grows gradually and S_2 remains non-dominant, indicating a landscape composed of multiple local-sized clusters rather than a single integrated network. A sharp transition

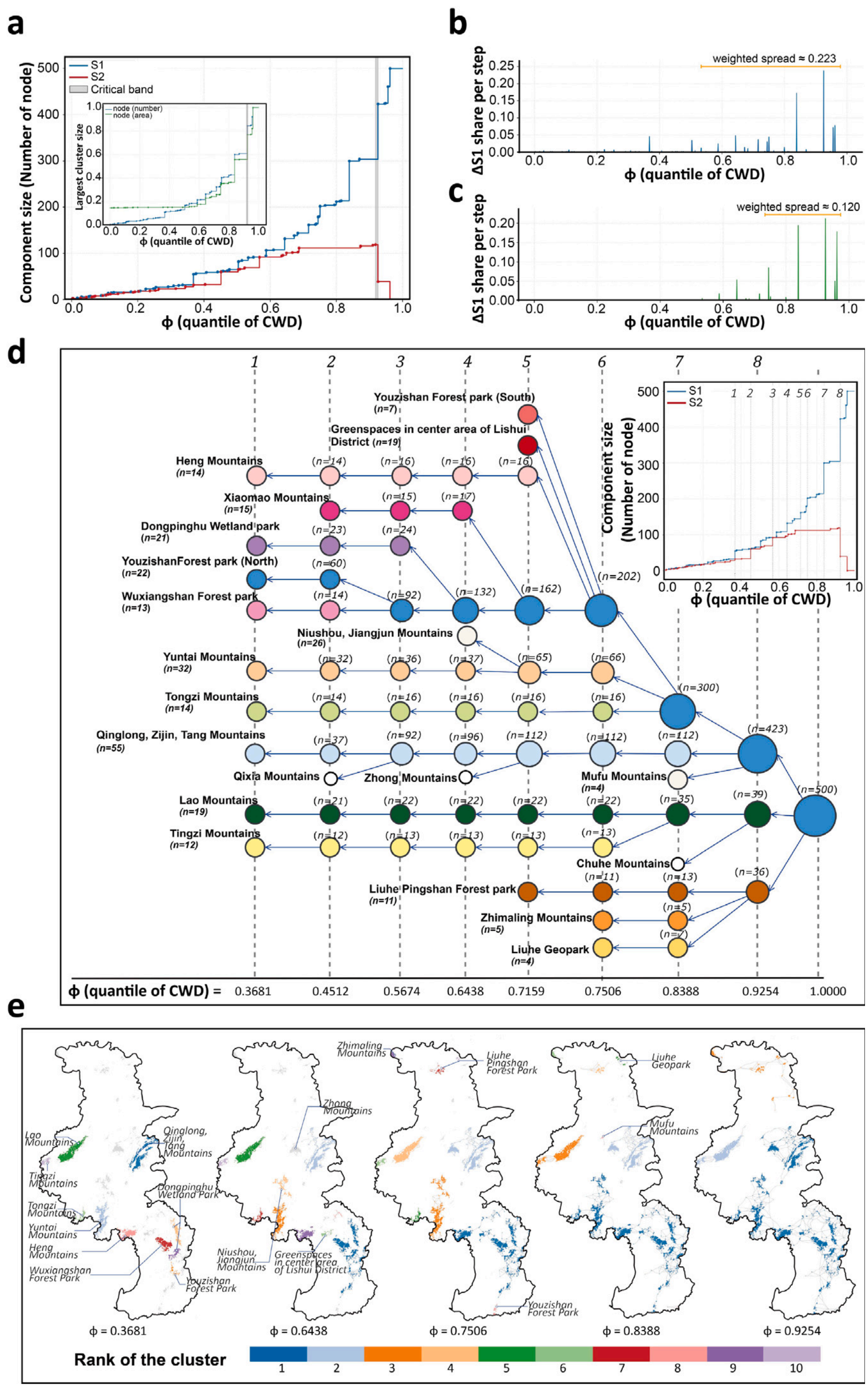

Fig. 4. Percolation scan: critical transition and hierarchical emergence. a, Evolution of the largest (S_1) and second–largest (S_2) components with quantile of CWD (ϕ). The gray band marks the critical band ($\phi = 0.9177$–0.9254) defined by the S_2 peak and collapse. The inset compares S_1 growth measured by patch count vs. patch area. **b–c,** Per–step gain of S_1 in count share (ΔS_1, weighted spread = 0.223) and in area share (ΔS_1, weighted spread = 0.120). The orange segment indicates the weighted mean ϕ of the transition plus/minus its weighted standard deviation, weighted by the positive per–step gains in ΔS_1 share. **d,** Bottom-up merger dendrogram labeling the major UGS clusters and their order of incorporation (small inset reproduces S-curves of count-based percolation for reference), full hierarchy in Fig. S5. **e,** Spatial snapshots at representative ϕ levels, visualizing the transition from many clusters to a single large component.

occurs within a narrow, high-cost critical band between $\phi = 0.9177$ and $\phi = 0.9254$. Within this band, S_2 reaches its maximum size and then collapses as it merges into S_1, while S_1 undergoes a rapid increase to form a system-spanning component. In other words, system-level connectivity is triggered only when a small set of very high-cost links is allowed to form, rather than emerging through gradual, citywide accretion.

The inset in Fig. 4a contrasts S_1 growth measured by patch count and by patch area. Both measures increase slowly at low and intermediate costs and then surge near the critical band. The area-based curve rises more steeply and with a slight delay relative to the count-based curve, suggesting that the decisive mergers involve a few large, area-dominant patches that join late along the cost axis. This size dependence is consistent with the per-step gains in S_1 (Fig. 4b–c): large increments in patch count and area are tightly concentrated

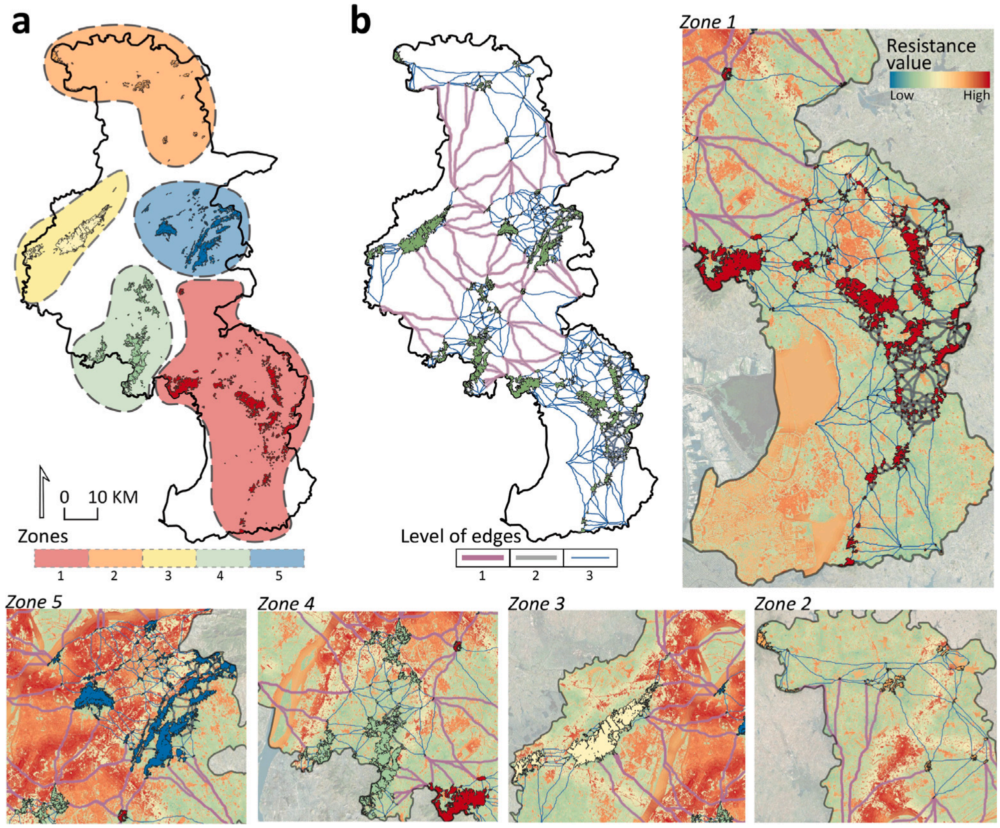

Fig. 5. Management zones and prioritized edges of the Nanjing UGS system. Integrating full percolation process with geographic regularization yields five management zones that respect major physical divides, and a three-level edge hierarchy whose Level 1 acts as cross-zone backbones. **a,** Final zoning (Zones 1–5) derived from geographically regularized consensus clustering. **b,** Edge hierarchy: Level 1 (inter-zonal edges), Level 2 (high-robustness intra-zonal edges), and Level 3 (remaining intra-zonal edges). Identification of Level 2 and Level 3 edges is based on a natural breaks classification of edge process robustness (R_{ij}). Details in Fig. S7. Zone-specific zoom-ins showing how Level 1 edges traverse high-resistance areas to connect different zones and the edges within each zone.

immediately before and within the critical band. The weighted spread of these gains is narrower for area (spread = 0.120) than for patch count (spread = 0.223), indicating that area consolidation is even more localized along the cost axis. For the emergence analysis below, we use count-based percolation as the primary descriptor of assembly history.

4.3.2. Structural emergence

The structural emergence of the UGS network is captured jointly by the merger dendrogram and spatial snapshots in Fig. 4d–e. The dendrogram (Fig. 4d) shows long intervals of incremental growth interspersed with a small number of major jumps, indicating that system-level connectivity is assembled through a limited sequence of large mergers rather than many small accretions. The spatial snapshots (Fig. 4e) connect these jumps to geography. At moderate cost levels (ϕ = 0.3681–0.7506), connectivity is organized into several well-defined local-sized clusters, often anchored by mountain ridges and large forest parks. As ϕ approaches and passes the critical band (ϕ = 0.8388 and ϕ = 0.9254), these clusters expand and are absorbed into the largest component, and the pattern stabilizes into three dominant subsystems: (i) a large, internally well-connected and system-spanning cluster south of the Yangtze River, (ii) a western cluster centered on the Lao Mountains, and (iii) a northern cluster around Liuhe District. Even after the network becomes system-spanning in graph terms, these subsystems remain recognizable in space, reflecting the persistent separating effects of the Yangtze River and the highly impervious urban core.

Taken together, Fig. 4d–e show that Nanjing's system-level connectivity emerges through a path-dependent merger sequence among a small number of geographically distinct subsystems. This ordered emergence pattern motivates the subsequent management classification of zones, edges, and nodes.

4.4. Ecological management mapping

Based on the percolation scan, we derive three management layers for Nanjing's UGS network — zones, a prioritized edge hierarchy, and a functional node typology — and combine them into an integrated ecological management map. Together, these outputs translate the network's assembly history into spatial priorities that are directly interpretable at the city scale.

4.4.1. Zones

The geographically regularized consensus clustering yields five ecologically coherent zones (k = 5) for Nanjing (Fig. 5a). The selected regularization parameter is λ = 0.15. The joint selection of k and λ is based on sensitivity analysis balancing functional cohesion and spatial contiguity; evaluation curves and intermediate partitions are summarized in Fig. S6. Each zone therefore groups patches with similar co-occurrence behavior across the full percolation process while remaining spatially coherent.

The zoning aligns with the three dominant subsystems identified in the emergence analysis (Fig. 4e, ϕ = 0.9254), while resolving their internal structure. Zones 1, 4, and 5 refine the large subsystem south of the Yangtze River into three functionally distinct parts: a southern mountain–wetland belt (Zone 1), a south-western foothill cluster (Zone 4), and an eastern mountain arc (Zone 5). Zone 3 corresponds to the Lao Mountains subsystem in the west, and Zone 2 captures the northern subsystem around Liuhe District. In practical terms, the five zones provide citywide management zones within which the edge hierarchy and node typology can be interpreted consistently.

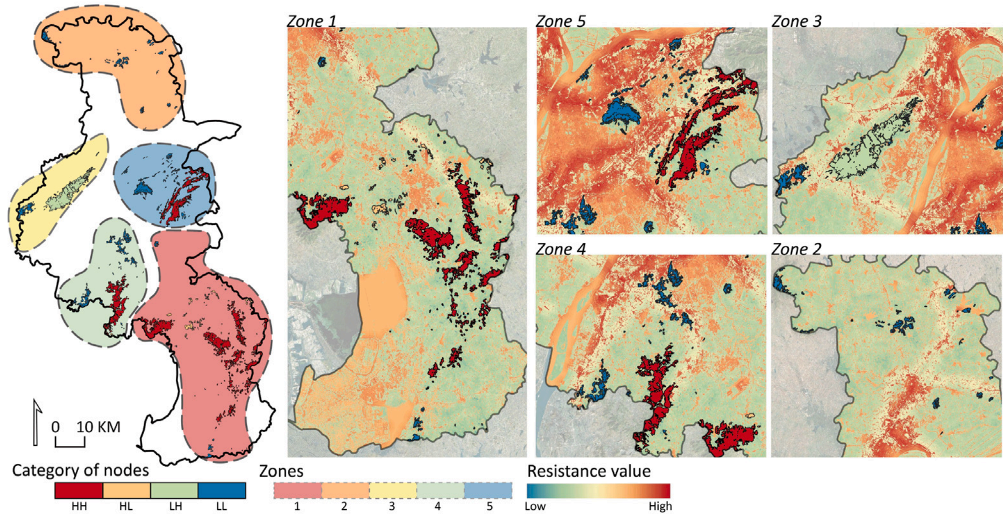

Fig. 6. Node functionality of the Nanjing UGS system. Nodes stratify into four functional categories in standardized (S_i, R_i) space: HH $(z(S_i) \geq 0,\ z(R_i) \geq 0)$, LH $(z(S_i) < 0,\ z(R_i) \geq 0)$, HL $(z(S_i) \geq 0,\ z(R_i) < 0)$, and LL $(z(S_i) < 0,\ z(R_i) < 0)$; details in the quadrant analysis (Fig. S7).

4.4.2. Edges

The edge hierarchy follows the three-level scheme (Fig. 2b). In Nanjing, this classification partitions the 1179 edges into 56 Level 1, 447 Level 2, and 676 Level 3 edges.

Level 1 edges are the only links between zones and therefore form the cross-zone backbone of the UGS network. They are relatively long, occur at high cost levels, and frequently traverse major barriers such as the Yangtze River and the highly impervious urban core (Fig. 5b). Because alternative cross-zone routes are scarce, the loss or degradation of even a small subset of Level 1 edges would quickly undermine system-level connectivity.

Level 2 edges represent robust intra-zonal structure. They are selected as the top two classes in a five-class natural breaks (Jenks) classification of R_{ij} (Fig. S7a). Spatially, Level 2 edges are strongly concentrated in the southern, mountain-dominated zones (Zones 1, 4, and 5), where they knit together hills and forest parks into dense internal networks. By contrast, Zone 2 (Liuhe) lacks Level 2 edges, and Zone 3 (Lao Mountains) contains only a modest number. This uneven distribution indicates marked differences in within-zone robustness across the city.

Level 3 edges comprise the remaining intra-zonal connections. They form a fine-grained local mesh that supports short-range permeability and adds redundancy around the more critical Level 1 and Level 2 structures. While any single Level 3 edge is rarely decisive for system-level connectivity, their collective presence can buffer local connectivity against small-scale disturbances, particularly where UGS is embedded in dense urban fabric.

4.4.3. Nodes

The node typology classifies all 500 UGS patches into four functional types based on the sign structure of standardized S_i and R_i (Fig. 2c). Interpreted as relative roles within the Nanjing network, this yields 234 HH nodes, 5 LH nodes, 56 HL nodes, and 205 LL nodes (Fig. 6) and captures how different patches contribute to connectivity over the entire percolation process (Fig. S7b–e).

HH nodes are dominated by large, contiguous forest–park complexes concentrated in Zones 1 and 5, with additional clusters in Zone 4. These patches are consistently embedded in large components throughout the scan and form stable structural cores that support high-capacity connectivity in the southern part of the city.

LH nodes are rare but strategically located, mostly near interfaces between zones. Their profile — frequent membership in large components despite weak and shifting co-membership — suggests a stepping-stone role that can facilitate cross-zone integration relative to their local cohesiveness.

HL nodes are mainly found in Zone 1, especially along the urban fringe. These patches are well integrated within local groups but join the system-spanning structure only at relatively high cost levels. They therefore support within-zone cohesion and redundancy more than early, citywide integration.

LL nodes are prevalent in Zone 2 and also appear as small, isolated green spaces on the fringes of other zones. These patches rarely participate in large components and contribute little to the core connectivity structure, highlighting areas where UGS exists but remains structurally peripheral from a network perspective.

4.4.4. Integrated management map

The integrated management map (Fig. 7a) overlays zones, edge levels, and node types into a single citywide representation of Nanjing's process-based UGS structure. The zone layer defines the primary management zones, the edge hierarchy identifies cross-zone backbones and robust within-zone cores, and the node typology highlights where structurally central patches (HH) and cross-zone stepping-stones (LH) are located relative to these corridors. Read together, these layers indicate where system-level connectivity is concentrated, where it is supported primarily by within-zone structure, and where fragmented areas are dominated by peripheral patches. This synthesis provides a direct spatial basis for prioritizing protection and reinforcement actions, which we relate to official planning in discussion (Section 5.1).

4.5. Robustness experiments

We test how well the percolation-based management map captures structurally important components by simulating progressive removal of edges and nodes and tracking the decline of weighted global efficiency E as an indicator of system-level connectivity (Fig. 8). The five-rule design brackets two planning-relevant scenario families—targeted loss of map-prioritized components (Map-attack) and delayed removal of those same components (Map-protect)—and benchmarks them against static-topology, ecological-function, and random removal rules. The comparison is reported both as full decline curves (Fig. 8) and as summary metrics (Table 3) to make differences in robustness more directly comparable.

At the edge level (Fig. 8a), Map-protect produces the slowest efficiency decline among the tested rules, with the highest AUC_E (0.538), the highest $E@20\%$ (0.897), and the largest $f_{E<0.5}$ (0.556). Relative

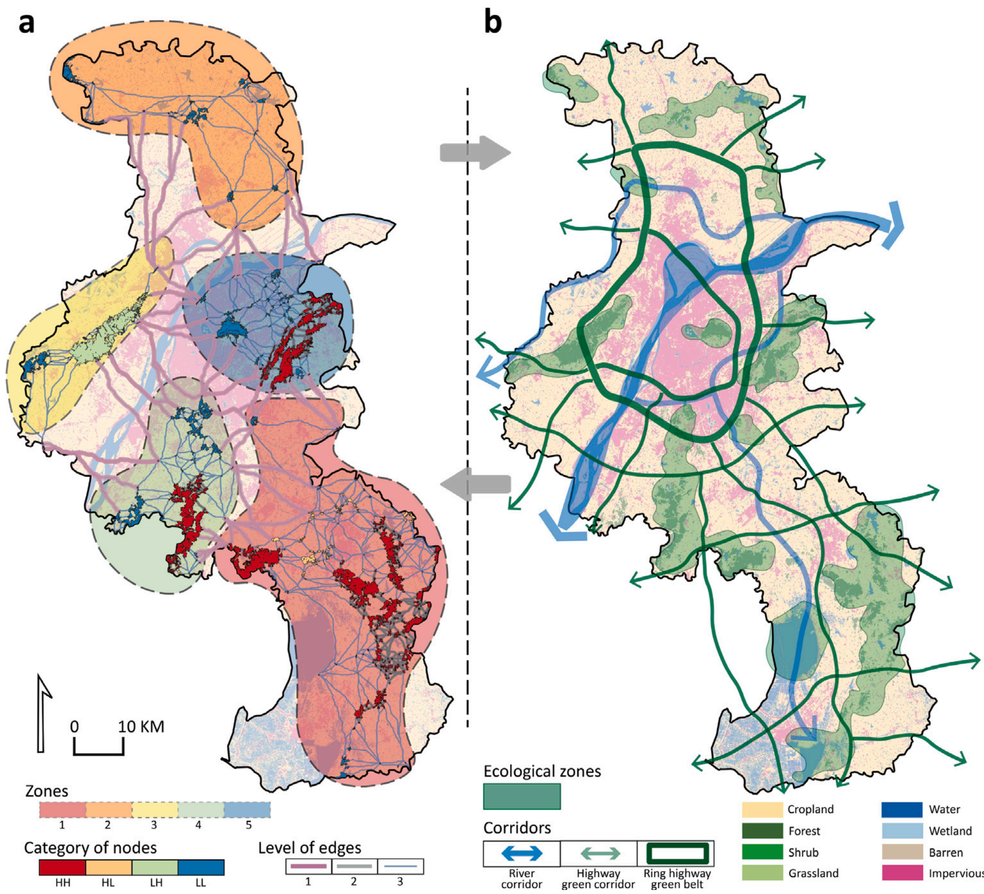

Fig. 7. Ecological management map versus official plans. Comparison between the percolation-based integrated management map (a) and the official, static-pattern-based "Ecological Security Structure" from Nanjing's 14th Five-Year Plan (b).

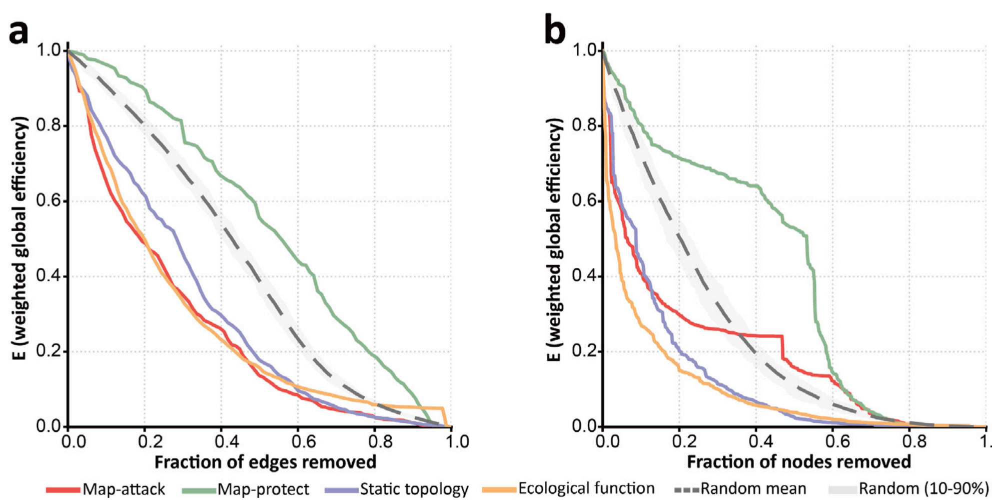

Fig. 8. Robustness experiments of system-level connectivity under edge and node removal. Weighted global efficiency E is normalized to the intact network and tracked as a function of the fraction of components removed. Curves show the five removal rules defined in Table 2; the dashed gray line is the random mean and the gray ribbon is the 10%–90% interval across random runs. **a,** Edge-based experiments. **b,** Node-based experiments.

to the random mean, it retains markedly more efficiency after 20% removal (0.897 vs. 0.802) and delays the drop below half-efficiency from 42.8% to 55.6% of removed edges. In contrast, Map-attack reduces E rapidly, reaching $E@20\% = 0.490$ and crossing the half-efficiency threshold after only 18.9% of edges are removed. The ecological-function baseline produces a similarly steep early decline, whereas the random mean and static-topology curves retain higher efficiency over the same early removal range. Taken together, these patterns indicate that the map-derived edge hierarchy is especially informative for identifying corridors whose protection sustains system-level connectivity and whose loss can quickly erode it.

At the node level (Fig. 8b), the clearest signal also appears in protection mode. Map-protect yields the highest AUC_E (0.404), the highest

Table 3
Summary metrics for the robustness experiments. AUC_E is the area under the normalized weighted global-efficiency curve, $E@20\%$ is the remaining efficiency after 20% of components have been removed, and $f_{E<0.5}$ is the removal fraction at which E first falls below 0.5. Higher values indicate slower efficiency loss under the corresponding removal rule.

Component	Removal rule	AUC_E	$E@20\%$	$f_{E<0.5}$
Edge	Map-attack	0.253	0.490	0.189
Edge	Map-protect	0.538	0.897	0.556
Edge	Static topology	0.298	0.616	0.284
Edge	Ecological function	0.271	0.499	0.199
Edge	Random mean	0.426	0.802	0.428
Node	Map-attack	0.192	0.298	0.064
Node	Map-protect	0.404	0.715	0.532
Node	Static topology	0.127	0.202	0.087
Node	Ecological function	0.098	0.150	0.034
Node	Random mean	0.244	0.504	0.203

$E@20\%$ (0.715), and the largest $f_{E<0.5}$ (0.532), indicating that postponing the removal of HH and LH nodes substantially delays the loss of system-level efficiency. Compared with the random mean, this rule preserves much more efficiency after 20% removal (0.715 vs. 0.504) and more than doubles the removal fraction needed to push E below 0.5 (0.532 vs. 0.203). The targeted-attack results are more nuanced: the ecological-function dPC baseline and static-topology betweenness baseline produce the steepest node-removal declines, while Map-attack remains more disruptive than the random mean. We therefore interpret the node typology primarily as a planning-oriented protection guide under the tested indicator, rather than as a strongest node-attack ranking across all possible benchmarks. Across both edges and nodes, the robustness analysis supports the map most strongly as a protection-oriented prioritization tool, while the attack curves reveal complementary rather than identical notions of structural importance.

5. Discussion

5.1. Planning implications and alignment with official planning

We translate the empirical outputs into implementation-oriented planning implications and compare them with the official "Ecological Security Structure" (Fig. 7). At a coarse scale, both representations highlight the hills-and-wetlands complex south of the Yangtze River, the Lao Mountains area, and the Liuhe cluster, indicating broad agreement on the main ecological cores. The percolation-based map adds operational detail by explicitly locating cross-zone backbones (Level 1), robust within-zone structure (Level 2), and stepping-stone patches (LH) that support system-level integration under cost constraints.

Three implication types follow. *Corridor safeguarding* focuses on Level 1 edges and their associated HH/LH nodes, which identify cross-zone connections that concentrate system-level dependence. These locations are natural candidates for stricter protection, regulation of encroachment, and targeted restoration where cross-barrier segments are weak. *Within-zone reinforcement* applies to zones with limited robust intra-zonal structure, particularly Zone 2, where the absence of Level 2 edges and the prevalence of LL nodes indicate low local redundancy. In such areas, patch expansion, new local links, or resistance reduction through land-use adjustment may be needed before system-level benefits can be expected. *Leverage-point activation* targets rare LH nodes near zone interfaces, where comparatively small interventions may yield disproportionate gains in cross-zone integration.

Overall, the percolation-based map complements the official plan by retaining the main spatial logic while making explicit which corridors and patches are most tightly associated with the emergence and maintenance of system-level connectivity under a cost-ordered assembly process. This distinction is important in relation to the broader connectivity literature. Classical graph and cost-based approaches have been highly successful in identifying ecological cores, potential corridors, and overall levels of connectivity (Adriaensen et al., 2003; McRae, 2006; Saura & Pascual-Hortal, 2007; Urban & Keitt, 2001), but they typically leave open the planning question of which locations should be protected first, reinforced within zones, or treated as cross-barrier leverage points. The present framework does not replace those approaches; instead, it reorganizes their diagnostic strengths into an intervention-oriented structure by separating management zones, corridor hierarchy, and node roles within a single process-based map.

5.2. Key insights from the percolation-based analysis in nanjing

Beyond the case-specific management implications, we interpret the main empirical patterns revealed by the Nanjing application, focusing on what the cost-ordered assembly process implies about the structure and vulnerability of system-level UGS connectivity.

5.2.1. A narrow high-cost transition and size-dependent mergers

The first insight is that system-spanning connectivity in Nanjing emerges through a narrow, high-cost transition that is size-dependent. Across most cost quantiles (ϕ), S_1 grows gradually while S_2 remains non-negligible (Fig. 4a), indicating that the system is dominated by multiple local-sized clusters rather than a single connected structure. Only within the high-cost critical band does S_2 peak and collapse into S_1, triggering the emergence of a system-spanning component. The area-based dynamics further indicate that a small number of large, area-dominant patches are incorporated late in the process (Fig. 4b–c), suggesting that a limited set of structurally pivotal elements controls the onset of system-level connectivity under cost constraints.

Viewed against recent percolation-oriented studies, this pattern supports the idea that urban connectivity can hinge on narrow transition intervals rather than change smoothly across the entire cost spectrum (Behnisch et al., 2019; Bian et al., 2023; Habrich & Fahrig, 2025). What the Nanjing case adds is a concrete planning interpretation of that transition: the decisive stage is not simply a theoretical threshold, but the moment when a small number of costly, barrier-crossing links reorganize the network from clustered subsystems into a city-scale structure.

5.2.2. Three spatial subsystems and corridor-mediated integration

The second insight is that, even after connectivity becomes system-spanning, the network remains organized into three spatial subsystems separated by persistent barriers. The emergence hierarchy and snapshots indicate a dominant southern subsystem, a western subsystem around the Lao Mountains, and a northern subsystem around Liuhe District (Fig. 4e). Their integration depends on a small set of high-cost cross-barrier connections, consistent with the limited number of inter-zonal Level 1 edges in the edge hierarchy (Fig. 5b). This pattern suggests that, in barrier-structured cities, system-level connectivity may depend disproportionately on a small number of cross-barrier corridors rather than on diffuse increases in local permeability.

This finding also clarifies how the present framework relates to LCP and other resistance-based connectivity analyses. Cost-based methods are well suited to identifying plausible routes through heterogeneous landscapes, but they do not by themselves indicate whether the most consequential interventions should focus on many local improvements or on a small set of cross-barrier bottlenecks (Aleixo et al., 2024; Balbi et al., 2021; Sawyer et al., 2011). In Nanjing, the process perspective shows that the latter issue is critical: cross-zone integration is governed by a very limited corridor set, whereas much of the remaining network contributes primarily to within-zone cohesion and redundancy.

5.2.3. Process-based mapping isolates leverage points for system-level connectivity

The third insight concerns the link between the process-based classification and system-level performance. The zones capture persistent co-occurrence structure across the full scan, while the edge and node classifications isolate elements that repeatedly participate in large structural components. This relationship becomes clearer when the map-based priorities are evaluated against static-topology, ecological-function, and random baselines using the common summaries AUC_E, $E@20\%$, and $f_{E<0.5}$ (Fig. 8; Table 3). Under this criterion, protecting map-prioritized edges and nodes yields the slowest decline of E among the tested rules, whereas targeted attack scenarios show that dPC and betweenness can capture additional notions of node criticality. We therefore interpret the map as a planning-oriented synthesis of structural leverage points, rather than as a ranking that dominates all possible connectivity metrics or disturbance scenarios.

This interpretation helps position the contribution of the study more precisely. The framework is strongest where planning requires spatial prioritization under incomplete information: it provides an operational way to move from diagnosis to differentiated action, while still allowing comparison against established topological and ecological-function benchmarks. Consistent with the gap summarized in Table 1, the main contribution is not a claim that percolation-derived rankings dominate every alternative baseline, but that they offer a coherent way to translate process-based emergence into management categories that can be read directly on the landscape (Bian et al., 2023; Habrich & Fahrig, 2025; Saura & Pascual-Hortal, 2007; Urban & Keitt, 2001).

5.3. Limitations, transferability, and future research

The framework is intended as a transferable workflow rather than a fixed parameter set. The same sequence of patch delineation, resistance modeling, cost-weighted connection construction, percolation scanning, and management mapping can be applied to other cities when the minimum data and modeling inputs are available (Table S4). What requires local calibration is the ecological interpretation of those inputs: patch definitions, resistance weights, linkage distances, transition criteria, mapping parameters, and optional validation or scenario data. In the Nanjing application, the resistance surface represents composite movement cost rather than species-specific or seasonally varying conditions, and the LCP-derived edge set may underrepresent diffuse or multiple-path movement where ecological flow is not concentrated along one dominant route (McRae et al., 2008; Sawyer et al., 2011; Zeller et al., 2024). The resulting map should therefore be interpreted as potential structural connectivity under the adopted resistance assumptions, with node and edge classes read relative to the focal network.

Future research can build on this workflow by enriching the ecological and planning assumptions around it. Species- or season-specific resistance scenarios could test how priorities shift with dispersal ability, barrier response, vegetation condition, hydrological state, or temporary changes in landscape permeability. Flow-based methods such as circuit theory could be coupled with the percolation scan to compare single-path and multiple-path representations of connectivity (Hall et al., 2021; Molné et al., 2023). Additional extensions could incorporate patch quality, habitat condition, governance constraints, and future land-use or climate scenarios. Comparative applications across multiple cities would further clarify which process-based priorities remain stable across local calibrations and which depend on context.

6. Conclusions

This study develops a process-based framework that operationalizes percolation theory for UGS connectivity assessment and ecological management mapping. Applied to a cost-weighted LCP network, the framework reconstructs how connectivity assembles as increasingly costly corridors become available and translates the full assembly history into three linked management outputs: management zones, a prioritized edge hierarchy, and a functional node typology. Evaluated with weighted global efficiency, these outputs identify protection-oriented priorities whose preservation can substantially delay the loss of system-level connectivity under the tested removal rules.

In Nanjing, the analysis reveals a narrow, high-cost critical band in which a small number of barrier-crossing links trigger the emergence of a system-spanning network. Even after this transition, the city remains organized into geographically distinct subsystems, indicating that system-level integration depends on a limited corridor set rather than on uniformly distributed local permeability. The resulting integrated map clarifies where cross-zone backbones, robust within-zone structure, persistent cores, and stepping-stone patches are located, providing a direct spatial basis for corridor safeguarding, within-zone reinforcement, and leverage-point protection.

The framework is transferable with local calibration rather than automatic parameter transfer. Its classifications should be interpreted relative to the focal system and aligned with local ecological assumptions, planning objectives, and data quality. Used in this way, percolation-based management mapping offers a concise route from dynamic connectivity diagnosis to spatial priorities that can support urban ecological planning across barrier-structured landscapes.

CRediT authorship contribution statement

Zixuan Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Sihao Gu:** Writing – review & editing, Investigation. **Horacio Samaniego:** Writing – review & editing, Investigation, Conceptualization. **Diego Rybski:** Investigation, Conceptualization. **Martin Behnisch:** Writing – review & editing, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.scs.2026.107564.

Data availability

Data will be made available on request.

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